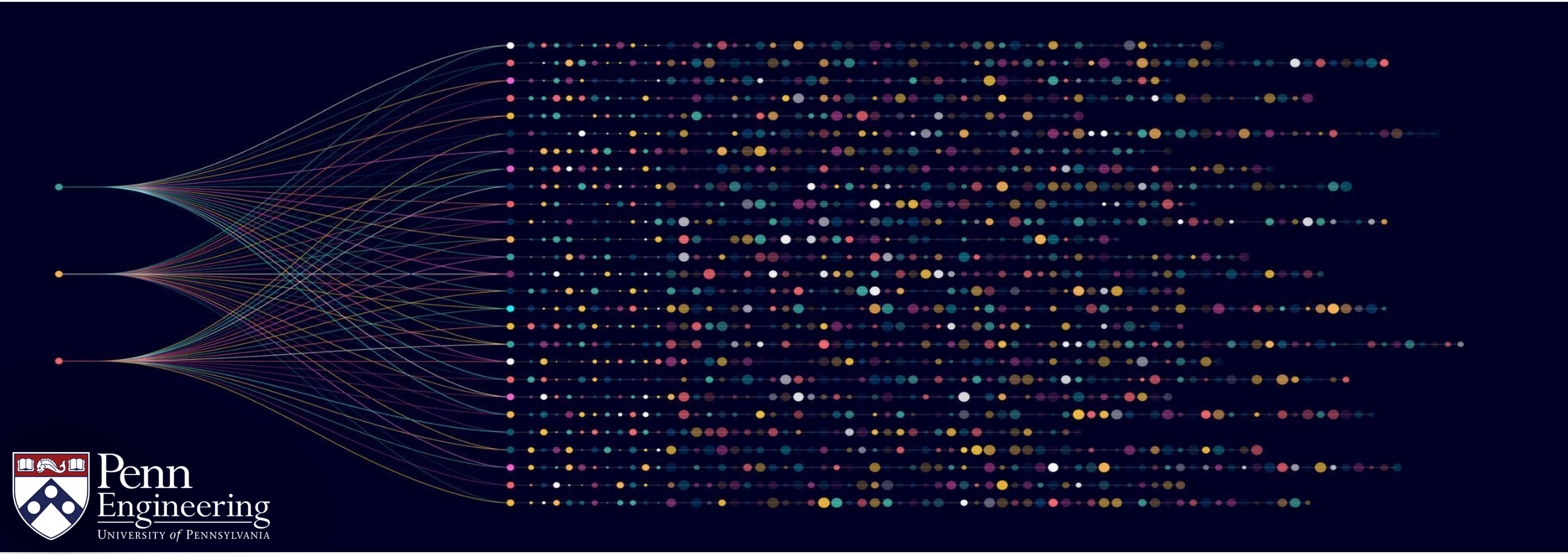
Algorithms for Big Data

Anindya De   Erik Waingarten

Introduction to linear algebra



Data in real life – exploring relationships

- A main goal of data analysis is to summarize a large amount of data in a meaningful manner and infer and understand relationships between various aspects of the data.

- This is difficult because of the large amount of data (“big data”) and complexity of each data point (“high-dimensional data”).

Data in real life – exploring relationships

- One of the main tools in our arsenal is that of “linear algebra.” In a nutshell, it will help us explore linear relationships in the data.

- This might seem like a limitation – “what about nonlinearity?”.
- But as we’ll see, exploring linear relationships is itself incredibly powerful! So, let’s dive in.

Some (synthetic) housing data

Obviously, the house price is related to the other attributes – but what exactly is the relationship?

Price (in thousands)	Square Feet	Number of Bedrooms	Number of Bathrooms	Age of the House (years)	Location Score	School District Score
687.2	1998.5	3.4	1.4	19.4	1.3	8.8
983.1	3842.3	1.6	2.5	13.6	6.7	6.6
770.9	3142.4	2.2	1.1	41.4	3.8	4.0
745.4	2715.7	2.5	3.7	17.8	5.6	1.6
462.7	1299.3	2.8	1.8	14.0	9.2	3.8
535.7	1299.2	4.1	3.0	27.1	3.2	3.9
425.1	985.9	1.8	1.9	7.0	4.7	7.6
971.8	3571.8	3.1	2.6	40.1	7.8	6.7
877.8	2723.6	3.4	2.6	3.7	3.1	9.0
739.7	3065.8	1.2	1.6	49.3	1.7	5.2
407.3	865.9	3.4	3.9	38.6	3.6	2.1
1034.3	3903.7	1.7	3.3	9.9	2.5	7.4
956.5	3463.8	1.3	3.8	0.3	9.4	7.8
639.3	1479.5	4.8	3.7	40.8	8.3	6.1
632.6	1381.8	4.9	2.8	35.3	6.7	7.9
592.2	1386.9	4.2	3.8	36.5	8.8	5.4
529.8	1773.6	2.2	1.3	38.6	8.2	5.7
660.9	2479.2	1.4	1.6	3.7	2.7	4.8
615.6	2182.2	3.7	1.1	17.9	9.0	1.2
533.7	1731.9	2.8	2.0	5.8	5.9	2.0

Some (synthetic) housing data

Obviously, the house price is related to the other attributes – but what exactly is the relationship?

Price = 0.2 x Sq ft + 40 x Bed# + 25 Bath# - 0.8 x Age + 15 x loc

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Some (synthetic) housing data

Obviously, the house price is related to the other attributes – but what exactly is the relationship?

Price = 0.2 x Sq ft + 40 x Bed# + 25 Bath# - 0.8 x Age + 15 x loc

How can we discover the relationship between these features?

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Linear algebra – basics

- The "atoms" in linear algebra are vectors – i.e., elements of $\mathbb{R}^n$. In other words, each vector is a n-tuple of real numbers.

- Addition of vectors is entry-wise – In other words,

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ x_3 + y_3 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Linear algebra – basics

- The "atoms" in linear algebra are vectors – i.e., elements of $\mathbb{R}^n$. In other words, each vector is a n-tuple of real numbers.

- Multiplication of vectors by scalars is also entry-wise.

$$c \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} c \cdot x_1 \\ c \cdot x_2 \\ c \cdot x_3 \\ \vdots \\ c \cdot x_n \end{pmatrix}.$$

Linear algebra – basics

- Suppose we have m vectors $v^{(1)}, \dots, v^{(m)} \in \mathbb{R}^n$.
- Then $span(v^{(1)}, \dots, v^{(m)})$ is the set of vectors v which can be expressed as $\sum_i \alpha_i \cdot v^{(i)}$ for $\alpha_1, \dots, \alpha_m \in \mathbb{R}$.

- In other words, $span(v^{(1)}, \dots, v^{(m)})$ is the set of vectors which can be expressed as a linear combination of $v^{(1)}, \dots, v^{(m)}$.

Some (synthetic) housing data

- Each column $v^{(1)}, \dots, v^{(7)} \in \mathbb{R}^{20}$.

- Finding a linear relation between price and other attributes is the same as: Is $v^{(1)} \in span\left(v^{(2)}, \dots, v^{(7)}\right)$?

- In fact, we want to also explicitly find out coefficients $\alpha_2, \dots, \alpha_7$ such that

$$v^{(1)} = \alpha_2\, v^{(2)} + \dots + \alpha_7\, v^{(7)}.$$

$v^{(1)}$ Price (in thousands)	$v^{(2)}$ Square Feet	$v^{(3)}$ Number of Bedrooms	$v^{(4)}$ Number of Bathrooms	$v^{(5)}$ Age of the House (years)	$v^{(6)}$ Location Score	$v^{(7)}$ School District Score
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Fundamental algorithmic question

- Suppose we're given vectors $w^{(1)}, \ldots, w^{(m)} \in \mathbb{R}^n$ and a target vector $v \in \mathbb{R}^n$.

- **Algorithmic questions:**

1. Is $v \in span(w^{(1)}, \ldots, w^{(m)})$?
2. If yes, can we find coefficients $\alpha_1, \ldots, \alpha_m$ such that $v = \sum_{j=1}^{m} \alpha_j \, . w^{(j)}$?

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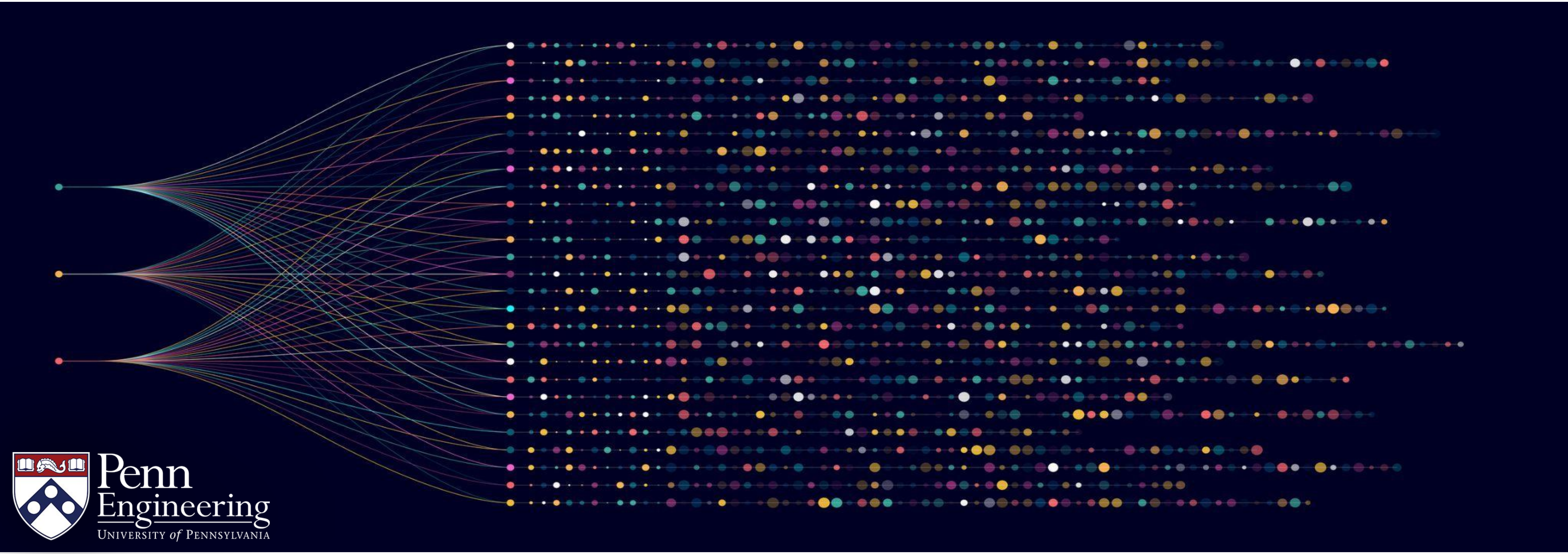
Gaussian elimination



Linear equations

- A fundamental computational problem is that of solving linear equations:

$$
\begin{aligned}
2x_1 + 3x_2 - x_3 &= 5 \\
4x_1 + x_2 + 2x_3 &= 6 \\
-2x_1 + 5x_2 + 7x_3 &= 3
\end{aligned}
$$

Gaussian elimination

- One of the most fundamental algorithms in computer science (arguably in all of science) is Gaussian elimination.

- In a nutshell, it is a procedure which given a system of linear equations, outputs a solution (or certifies that there is no solution).

Main idea: Gaussian elimination

- Main idea: Transform the system of equations into a simple form while preserving the solutions.

- In particular, we can add and subtract "scalar multiples" of equations and this will preserve the solution set.

Gaussian elimination: an example

$$\begin{aligned} 2x_1 + 3x_2 - x_3 &= 5 \\ 4x_1 + x_2 + 2x_3 &= 6 \\ -2x_1 + 5x_2 + 7x_3 &= 3 \end{aligned}$$

- Subtract 2 x Equation (1) from Equation (2).

Gaussian elimination: an example

$$\begin{aligned} 2x_1 \; &+ 3x_2 - \; x_3 &&= 5 \\ \color{red}{0}x_1 \; &-5x_2 + \; 4x_3 &&= -4 \\ -2x_1 \; &+ 5x_2 + 7x_3 &&= 3 \end{aligned}$$

- Add Equation (1) to Equation (3).

Gaussian elimination: an example

$$
\begin{aligned}
2x_1 &+ 3x_2 - x_3 = 5 \\
0x_1 &- 5x_2 + 4x_3 = -4 \\
0x_1 &+ 8x_2 + 6x_3 = 8
\end{aligned}
$$

- Add Equation (1) to Equation (3).

Gaussian elimination: an example

$$\begin{aligned} 2x_1 &+ 3x_2 - x_3 = 5 \\ 0x_1 &- 5x_2 + 4x_3 = -4 \\ 0x_1 &+ 8x_2 + 6x_3 = 8 \end{aligned}$$

- Add Equation (1) to Equation (3). Now the variable x_1 appears only in Equation (1).

Gaussian elimination: an example

$$
\begin{aligned}
2x_1 \; &+ 3x_2 - \; x_3 &&= 5 \\
0x_1 \; &-5x_2 + \; 4x_3 &&= -4 \\
0x_1 \; &+ 8x_2 + 6x_3 &&= 8
\end{aligned}
$$

- Next add ${}^{8}/_{5}$ x Equation (2) to Equation (3).

Gaussian elimination: an example

$$
\begin{aligned}
2x_1 + 3x_2 - x_3 &= 5 \\
0x_1 - 5x_2 + 4x_3 &= -4 \\
0x_1 + 0x_2 + {}^{62}/_5\, x_3 &= {}^{8}/_5
\end{aligned}
$$

- Now, notice that the third equation only contains variable x_3.

Gaussian elimination: an example

$$\begin{aligned} 2x_1 + 3x_2 - x_3 &= 5 \\ -5x_2 + 4x_3 &= -4 \\ {}^{62}/_{5}\, x_3 &= {}^{8}/_{5} \end{aligned}$$

- From this, we now get $x_3 = {}^{8}/_{62} = {}^{4}/_{31}$.

Gaussian elimination: an example

$$\begin{aligned} 2x_1 + 3x_2 - x_3 &= 5 \\ -5x_2 + 4x_3 &= -4 \\ {}^{62}/_5\, x_3 &= {}^{8}/_5 \end{aligned}$$

- With $x_3 = {}^{4}/_{31}$, we can now start back-propagating to get the other variables. In particular, Equation (2) gives us $x_2 = {}^{28}/_{31}$.

Gaussian elimination: an example

$$
\begin{aligned}
2x_1 + 3x_2 - x_3 &= 5 \\
-5x_2 + 4x_3 &= -4 \\
{}^{62}/_{5}\, x_3 &= {}^{8}/_{5}
\end{aligned}
$$

- Finally, since we have x_2 and x_3, from Equation (2), we have $x_1 = {}^{75}/_{62}$.

Gaussian elimination: in summary

- Given the three equations in variables x_1, x_2 and x_3, we do a sequence of elementary row operations so that
  the 3rd equation only involves variables $\{x_3\}$;
  the 2nd equation only involves variables $\{x_2, x_3\}$;
  the 1st equation only involves variables $\{x_1, x_2, x_3\}$;

More general system of linear equations

- A system of n linear equations in ℓ variables

$$x_1 \cdot w_1^{(1)} + x_2 \cdot w_1^{(2)} + \dots\dots + x_\ell \cdot w_1^{(\ell)} = b_1$$

$$x_1 \cdot w_2^{(1)} + x_2 \cdot w_2^{(2)} + \dots\dots + x_\ell \cdot w_2^{(\ell)} = b_2$$

$$\vdots$$

$$x_1 \cdot w_n^{(1)} + x_2 \cdot w_n^{(2)} + \dots\dots + x_\ell \cdot w_n^{(\ell)} = b_n$$

Gaussian elimination

- Guarantee of Gaussian elimination: given n equations in ℓ variables, it will run in time $\mathcal{O}(n\ell^2 + \ell n^2)$ and output whether the system has zero, one or infinite solutions.

- In the latter two cases, it will also output a solution. In fact, if the number of solutions is ∞, it will also describe how to obtain "all such solutions".

Existence of a non-zero solution: Rank-Nullity theorem

Consider the following system of equations.

$$
\begin{aligned}
x_1 \cdot w_1^{(1)} + x_2 \cdot w_1^{(2)} + \ldots \ldots + x_\ell \cdot w_1^{(\ell)} &= 0 \\
x_1 \cdot w_2^{(1)} + x_2 \cdot w_2^{(2)} + \ldots \ldots + x_\ell \cdot w_2^{(\ell)} &= 0 \\
&\vdots \\
x_1 \cdot w_n^{(1)} + x_2 \cdot w_n^{(2)} + \ldots \ldots + x_\ell \cdot w_n^{(\ell)} &= 0
\end{aligned}
$$

If the number of variables $\ell >$ the number of equations n, then there exists a non-zero solution to the above system.

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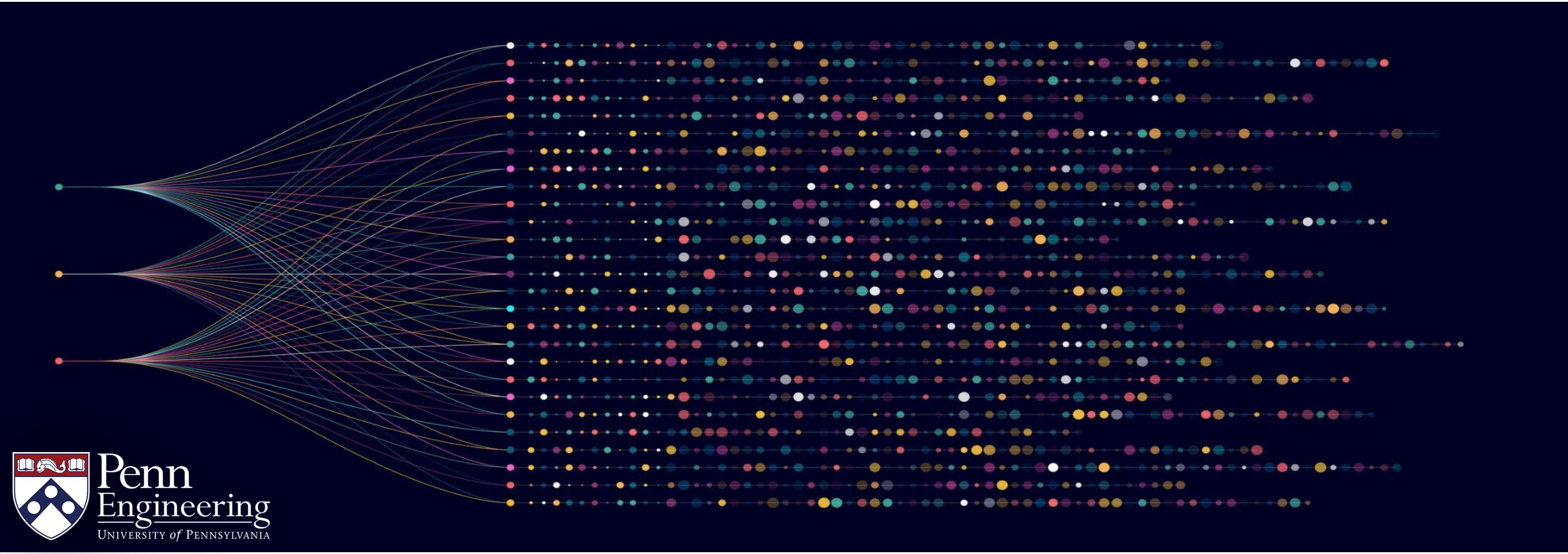
Computational applications of Gaussian elimination



Computing the span

- Suppose we're given vectors $v^{(1)}, \ldots, v^{(m)} \in \mathbb{R}^n$ and a target vector $w \in \mathbb{R}^n$.

- A basic algorithmic question: Is $w \in span(v^{(1)}, \ldots, v^{(m)})$?

Setting up a linear system

- We are looking to find coefficients $\alpha_1, \dots, \alpha_m \in \mathbb{R}$ such that a target vector $\sum_{j=1}^{m} \alpha_j \cdot v^{(j)} = w$.

- If the i^{th} coordinate of $v^{(j)}$ is denoted by $v_i^{(j)}$, then for $1 \leq i \leq n$,

$$\sum_{j=1}^{m} \alpha_j \cdot v_i^{(j)} = w_i. \qquad (*)$$

Setting up a linear system

- If the i^{th} coordinate of $v^{(j)}$ is denoted by $v_i^{(j)}$, then for $1 \leq i \leq n$,

$$\sum_{j=1}^{m} \alpha_j \cdot v_i^{(j)} = w_i. \qquad (*)$$

- Can be seen as n linear equations in the unknowns $\alpha_1, \dots, \alpha_m$.

- Apply Gaussian elimination in time $O(nm(n+m))$ to find out a feasible choice of $\alpha_1, \dots, \alpha_m$ (if one exists).

Computing the span: running time

- Thus given vectors $v^{(1)}, \ldots, v^{(m)} \in \mathbb{R}^n$ and a target vector $w \in \mathbb{R}^n$, we can use Gaussian elimination to decide if $w \in span(v^{(1)}, \ldots, v^{(m)})$ in time $O(nm(n+m))$.

- This can be used as a subroutine for many other algorithms in linear algebra – e.g., computing a basis for a subspace.

Computing the basis for a subspace

We are given a set of vectors $v^{(1)}, \dots, v^{(m)} \in \mathbb{R}^n$ and we want to find a linearly independent set B such that

$$span(B) = span\left(v^{(1)}, \dots, v^{(m)}\right).$$

Algorithm: Computing the basis for a subspace

1. We initialize B to the empty set.

2. Iterate through $i = 1$ to m $\{$

3. Check if $v^{(i)} \in span\,(B)$. If no, $B \leftarrow B \cup v^{(i)}$. $\}$

4. Output B.

At termination, B is a linearly independent set such that $span(B) = span(v^{(1)}, \ldots, v^{(m)})$.

Running time analysis

1. We initialize B to the empty set.

2. Iterate through $i = 1$ to m {

3. Check if $v^{(i)} \in span(B)$. If yes, $B \leftarrow B \cup v^{(i)}$.}

4. Output B.

Time complexity of Step 3 in the i^{th} iteration: $O(in(i+n))$.

Total time complexity: $\sum_{i=1}^{m} in(i+n) = O(n^2m^2 + nm^3)$.

Possible to optimize this to $O(n^2m + nm^2) = O(nm(n+m))$.

Summary

- Gaussian elimination is a fundamental subroutine in the design of algorithms in linear algebra.

- Many more applications – in later videos and assignments.

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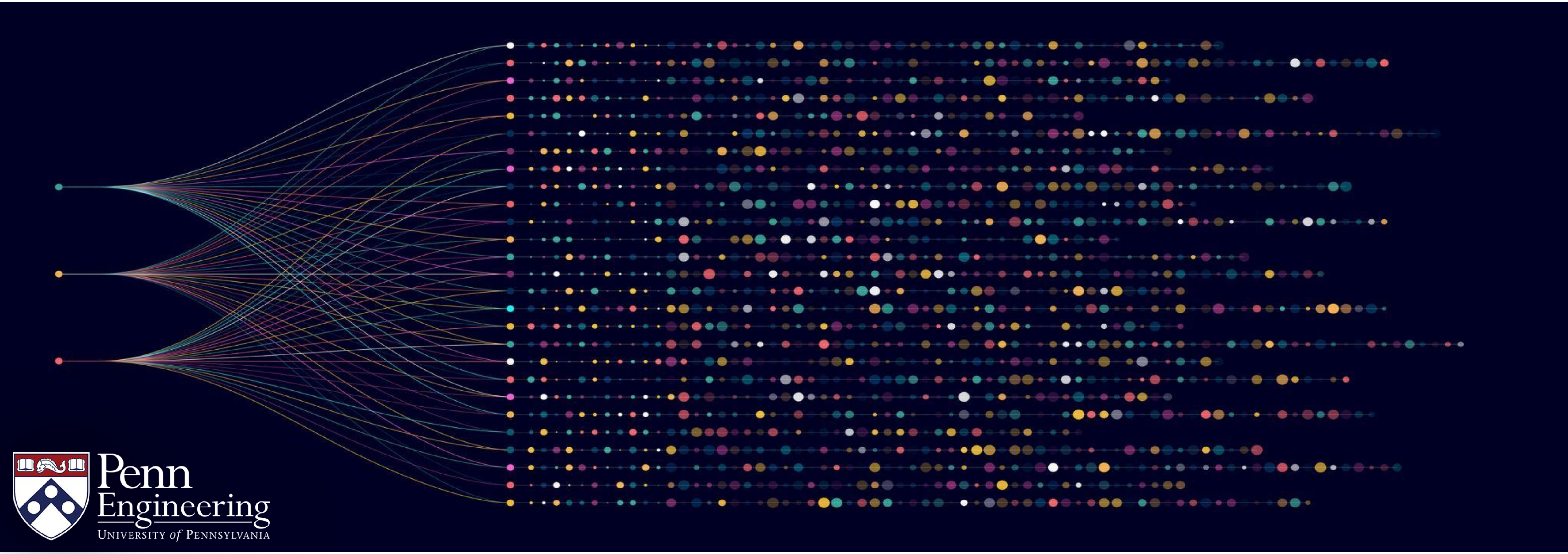
Linear dependence and independence



Linear dependence and independence

- A fundamental concept in linear algebra is that of linear dependence: Let $v^{(1)}, \dots, v^{(m)}$ be m vectors in $\mathbb{R}^n$.

- We say that $v^{(1)}, \dots, v^{(m)}$ are linearly dependent if there exists real numbers $\alpha_1, \dots., \alpha_m$ such that $\sum_i \alpha_i \cdot v^{(i)} = 0$ and not all $\alpha_1, \dots., \alpha_m$ are zero.

- Otherwise, $v^{(1)}, \dots, v^{(m)}$ are linearly independent.

Consequence of linear dependence

- Suppose $v^{(1)}, \dots, v^{(m)}$ are linearly dependent.
- That means, we have $\sum_i \alpha_i \cdot v^{(i)} = 0$.
- Suppose $\alpha_j \neq 0$ (such a coordinate exists).
- Then, $v^{(j)} = \sum_{i \neq j} \frac{-\alpha_i}{\alpha_j} \cdot v^{(i)}$.

Consequence of linear dependence

- More succinctly, $v^{(j)} \in span\left(v^{(1)}, \dots, v^{(j-1)}, v^{(j+1)}, \dots, v^{(m)}\right)$.
- We call such a vector $v^{(j)}$ as *linearly redundant* for the system $v^{(1)}, \dots, v^{(m)}$.

- **Observation:** If $v^{(j)}$ is *linearly redundant* for $v^{(1)}, \dots, v^{(m)}$, then

$$span\left(v^{(1)}, \dots, v^{(m)}\right) = span\left(v^{(1)}, \dots, v^{(j-1)}, v^{(j+1)}, \dots, v^{(m)}\right).$$

Iterating until we get linear independence

- Given a set of vectors $v^{(1)}, \ldots, v^{(m)}$, all in $\mathbb{R}^n$, we can consider the following process:

1. If the set is linearly independent, then halt.
2. If the set is linearly dependent, then let $v^{(j)}$ be a linearly redundant vector. Remove $v^{(j)}$ from the set and repeat.

Iterating until we get linear independence

When the process terminates, we are left with a set B with the following properties:

1. $B = \{v^{(i_1)}, \ldots, v^{(i_t)}\}$ where the set is linearly independent.
2. For all $1 \le j \le m$, $v^{(j)} \in span(B)$. Consequently, we have

   $span(B) = span\left(v^{(1)}, \ldots, v^{(m)}\right)$.

Iterating until we get linear independence

When the process terminates, we are left with a set B with the following properties:

1. $B = \{v^{(i_1)}, \dots, v^{(i_t)}\}$ where the set is linearly independent.
2. For all $1 \le j \le m$, $v^{(j)} \in span(B)$. Consequently, we have $span(B) = span(v^{(1)}, \dots, v^{(m)})$.

Such a set B is called a basis for $span(v^{(1)}, \dots, v^{(m)})$.

Existence of a basis

Thus, in summary, for any $v^{(1)}, \dots, v^{(m)}$, there is a linearly independent set B such that $span(B) = span\left(v^{(1)}, \dots, v^{(m)}\right)$.

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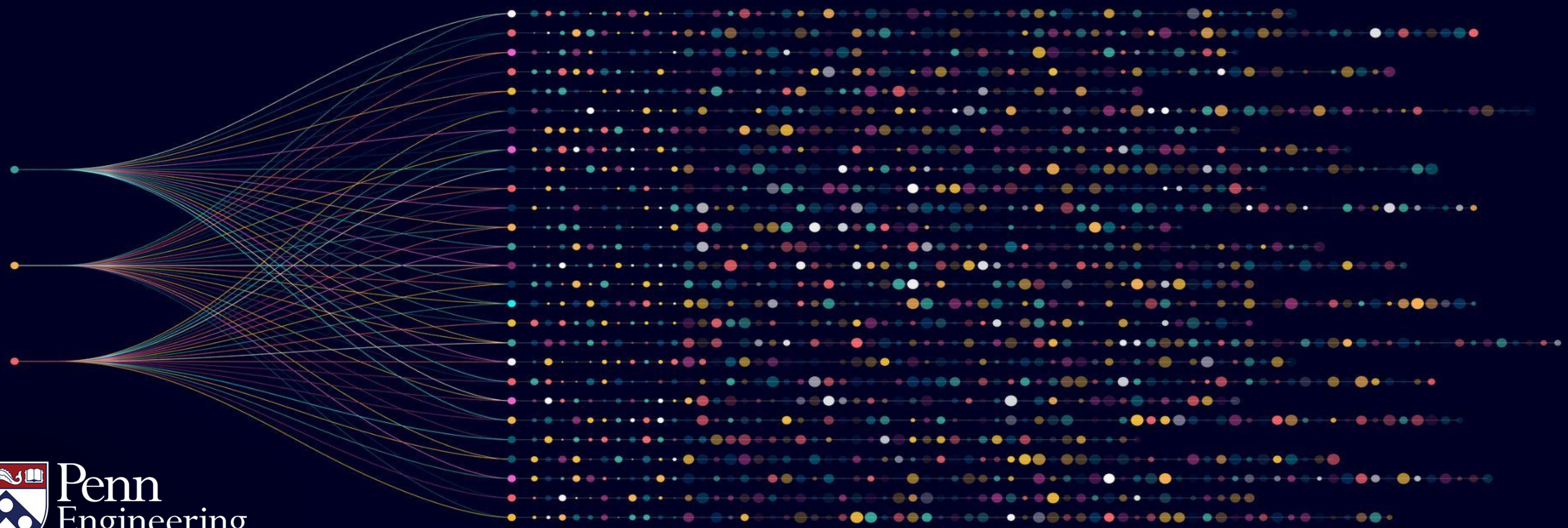
Such a set B is called a basis for $span\left(v^{(1)}, \dots, v^{(m)}\right)$.

Question: what about the computational aspect of computing a basis B? This will be addressed in a later video.

Algorithms for Big Data

Anindya De   Erik Waingarten

Rank of a matrix



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Column-rank of a matrix

Suppose we are given data in the form of m vectors – each in $\mathbb{R}^n$.

Let us call these vectors as $v^{(1)}, \dots, v^{(m)}$.

We arrange this data as a matrix $V \in \mathbb{R}^{nxm}$ -- it has m columns

where the i^{th} column is $v^{(i)}$.

The col-rank of V:= $\dim\left(span\left(v^{(1)}, \dots, v^{(m)}\right)\right)$.

Column-rank of a matrix: an example

$$V = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -1 & 1 \\ -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 5 & -2 & 1 \end{pmatrix}$$

Column-rank of a matrix: an example

$$V = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -1 & 1 \\ -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 5 & -2 & 1 \end{pmatrix}$$

The col-rank of matrix V is 3 because the columns $v^{(1)}, v^{(2)}, v^{(3)}$ are linearly independent but $v^{(4)} = v^{(2)} - 2v^{(3)} - v^{(1)}$.

Column rank vs row-rank

$$V = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -1 & 1 \\ -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 5 & -2 & 1 \end{pmatrix}$$

One can ask what happens if we define the analogous notion of row-rank?

Row-rank of a matrix

$$V = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -1 & 1 \\ -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 5 & -2 & 1 \end{pmatrix}$$

We can also think of V row-wise – let the rows be called $u_{[1]}, \dots, u_{[5]}$ – each of them an element of $\mathbb{R}^4$.

Rank of a matrix: rows vs columns

$$V = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -1 & 1 \\ -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 5 & -2 & 1 \end{pmatrix}$$

Row-rank:= rank of the space spanned by the rows of V.

Miracle from linear algebra!!

$$V = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 4 & -1 & 1 \\ -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 3 & 5 & -2 & 1 \end{pmatrix}$$

Surprising fact from linear algebra: The notion of rank remains unchanged whether we look at the matrix in terms of column vectors or row vectors. Often referred to as "row-rank = column-rank".

Factorization of matrices based on rank

Suppose $V \in \mathbb{R}^{n \times m}$ is a matrix with $rank(V) = k$. Let its columns be $v^{(1)}, \ldots, v^{(m)} \in \mathbb{R}^n$. This means there are k vectors $u^{(1)}, \ldots, u^{(k)} \in \mathbb{R}^n$ such that each $v^{(i)} \in span\left(u^{(1)}, \ldots, u^{(k)}\right)$.

Thus each $v^{(i)} = \sum_{\ell=1}^{k} w_{\ell i} u^{(\ell)}$ for real numbers $\{w_{\ell i}\}_{1 \le \ell \le k, 1 \le i \le m}$.

Factorization of matrices

Representing $v^{(i)} = \sum_{\ell=1}^{k} w_{\ell i} u^{(\ell)}$ by matrix-vector product.

$$v^{(i)} = \underbrace{\begin{bmatrix} u^{(1)} & u^{(2)} & \dots & \dots & u^{(k)} \end{bmatrix}}_{U} \begin{bmatrix} w_{1i} \\ w_{2i} \\ \vdots \\ \vdots \\ w_{ki} \end{bmatrix}$$

Factorization of matrices

Representing all of $v^{(1)}, \ldots, v^{(m)}$ by matrix-vector product.

$$[v^{(1)} \quad v^{(2)} \quad .. \quad v^{(m)}] = \underbrace{[u^{(1)} \quad u^{(2)} \quad \cdots \quad u^{(k)}]}_{U} \underbrace{\begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1i} \\ w_{21} & w_{22} & \cdots & w_{1i} \\ \vdots & \vdots & \vdots & \vdots \\ w_{k1} & w_{k2} & \cdots & w_{km} \end{bmatrix}}_{W}$$

Factorization of matrices

Representing all of $v^{(1)}, \dots, v^{(m)}$ by matrix-vector product.

$$\underbrace{\begin{bmatrix} v^{(1)} & v^{(2)} & .. & v^{(m)} \end{bmatrix}}_{V} = \underbrace{\begin{bmatrix} u^{(1)} & u^{(2)} & \dots & u^{(k)} \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1i} \\ w_{21} & w_{22} & \cdots & w_{1i} \\ \vdots & \vdots & \vdots & \vdots \\ w_{k1} & w_{k2} & \cdots & w_{km} \end{bmatrix}}_{W}$$

- Thus, a rank k matrix V can be expressed as UW where $U \in \mathbb{R}^{nxk}$ and $W \in \mathbb{R}^{kxm}$.

Factorization of matrices

- Thus, a rank k matrix V can be expressed as UW where $U \in \mathbb{R}^{nxk}$ and $W \in \mathbb{R}^{kxm}$.

- This means even though V has mn entries it can be expressed as a product of U (with nk entries) and W (with km entries).

- Thus a rank k matrix $V \in \mathbb{R}^{nxm}$ can be expressed using $nk + km$ real numbers.

Low-rank matrices

Rank of a matrix = measure of complexity.

Example: Suppose $V \in \mathbb{R}^{nxm}$ has rank $k = \Theta(1)$. Then, V can be represented using $k(n+m) = O(n+m)$ real numbers. Contrast this with the fact that V actually has nm entries.

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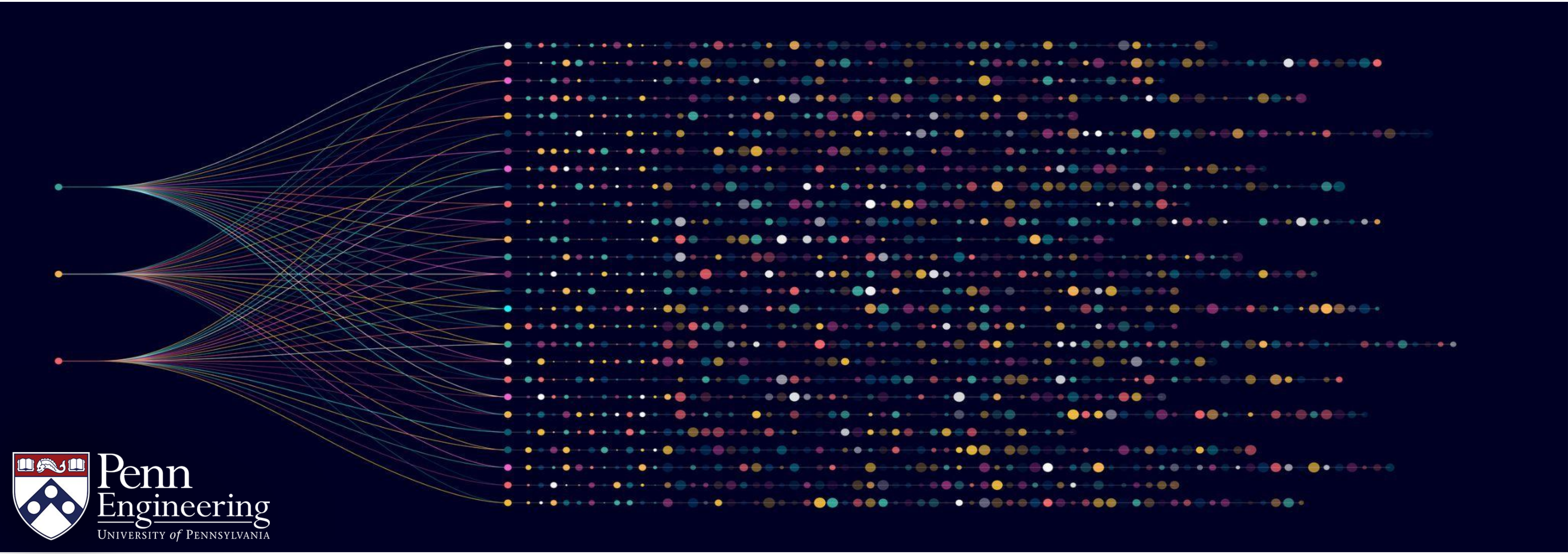
Some more basics of linear algebra



Subspace (generative notion)

A subset $V \subseteq \mathbb{R}^n$ is said to be a subspace if there are vectors $v^{(1)}, \ldots, v^{(m)} \in \mathbb{R}^n$ such that $V = span\left(v^{(1)}, \ldots, v^{(m)}\right)$.

For any subspace V, observe that for any real numbers α, β : If $x, y \in V$, then $\alpha x + \beta y \in V$.

Dual definition of subspaces

There is another "discriminative" definition of subspaces. Namely, a subset $V \subseteq \mathbb{R}^n$ is said to be a subspace if the following hold:

I. The element $\mathbf{0} \in V$.

II. For $x, y \in V$ and $\alpha, \beta \in \mathbb{R}$, $\alpha x + \beta y \in V$.

In other words, a subspace V is defined to be a subset of $\mathbb{R}^n$ which is closed under linear combinations and contains $\mathbf{0}$.

Equivalence of the two definitions

The two definitions of subspaces turns out to be equivalent.
In particular, if a set V satisfies the “discriminative” definition of subspaces, then $V = \operatorname{span}\left(v^{(1)}, \dots, v^{(m)}\right)$ for some $v^{(1)}, \dots, v^{(m)} \in \mathbb{R}^n$.

In the other direction, if $V = \operatorname{span}\left(v^{(1)}, \dots, v^{(m)}\right)$, then it satisfies the “discriminative” definition of subspaces.

Equivalence of the two definitions

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In particular, if a set V satisfies the “discriminative” definition of subspaces, then $V = \text{span}\left(v^{(1)}, \dots, v^{(m)}\right)$ for some $v^{(1)}, \dots, v^{(m)} \in \mathbb{R}^n$.

In the other direction, if $V = \text{span}\left(v^{(1)}, \dots, v^{(m)}\right)$, then it satisfies the “discriminative” definition of subspaces.

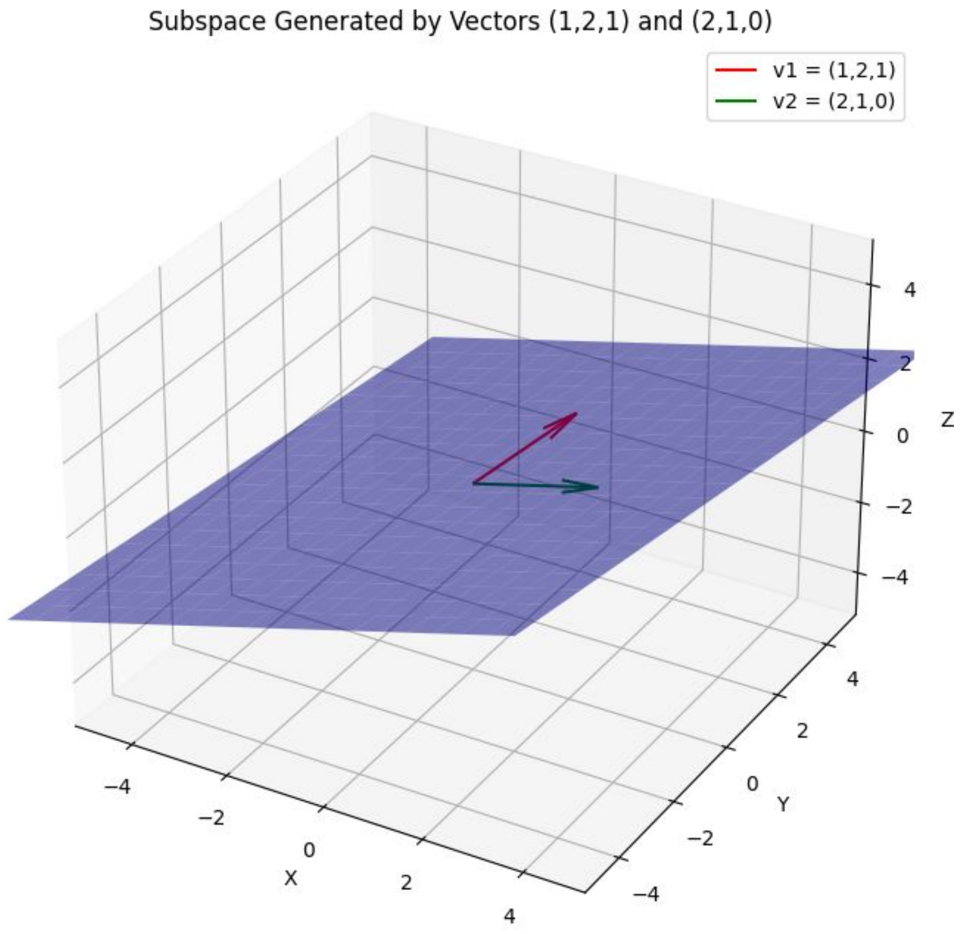
We skip the proof (both sides are easy – though the first uses the rank-nullity theorem).

Example of a subspace

Let $v^{(1)} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ and $v^{(2)} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$. Then, $V = span(v^{(1)}, v^{(2)})$ consists of

$$\alpha \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \alpha + 2\beta \\ 2\alpha + \beta \\ \alpha \end{pmatrix}, \text{ for all } \alpha, \beta \in \mathbb{R}.$$

Example of a subspace



Subspaces: basis and dimension

Suppose we are given a subspace $V = \text{span}(v^{(1)}, \ldots, v^{(m)})$.

It follows that there is a linearly independent set $w^{(1)}, \ldots, w^{(k)}$ such that $V = \text{span}(w^{(1)}, \ldots, w^{(k)})$.

Such a linearly independent set $\{w^{(1)}, \ldots, w^{(k)}\}$ is said to be a basis of V.

Subspaces: basis and dimension

Fact: Every basis for V has the same cardinality.

In other words, if both $\{w^{(1)}, \dots, w^{(k)}\}$ and $\{v^{(1)}, \dots, v^{(\ell)}\}$ are bases for V, then $k = \ell$.

Definition: The dimension of V (denoted by $dim(V)$) is the number of elements in a basis for V.

More facts about subspaces

We now list some handy facts about subspaces (we will skip the proofs for now but they can appear in the recitation/assignment).

Fact 1: Suppose V is a subspace and $v^{(1)} \dots, v^{(m)} \in V$ are linearly independent. If $m = \dim(V)$, then $v^{(1)} \dots, v^{(m)}$ is a basis for V.

If $m < \dim(V)$, then there is a basis of V $\{v^{(1)} \dots, v^{(m)}, w^{(1)}, \dots, w^{(k)}\}$ where $k = \dim(\mathrm{V}) - m$.

More facts about subspaces

Fact 2: If V is a subspace and $v^{(1)}, \dots, v^{(m)} \in V$ such that $m > \dim(V)$, then $v^{(1)}, \dots, v^{(m)}$ are linearly dependent -- i.e., for some $\alpha_1, \dots, \alpha_m$ (not all zero) such that $\sum_i \alpha_i \cdot v^{(i)} = 0$.

Algorithms for Big Data

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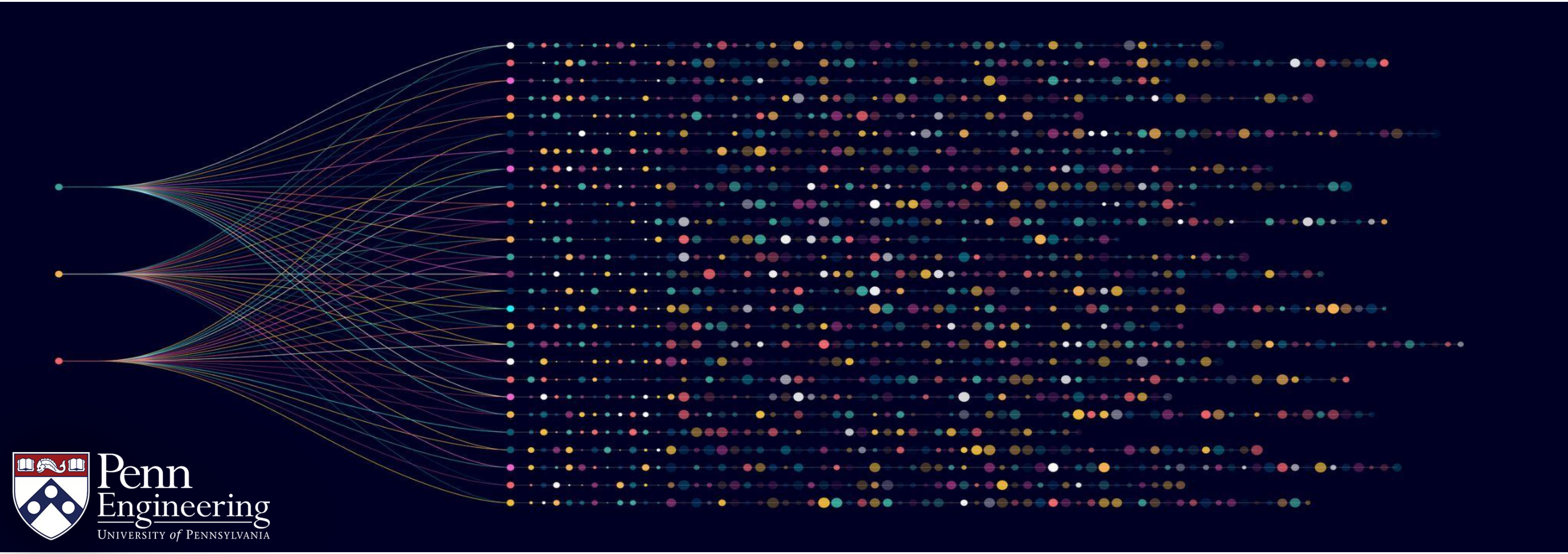
Some basics of matrices



Matrices

Matrices are rectangular array of numbers. A mxn matrix $A \in \mathbb{R}^{mxn}$ has m rows and n columns.

Examples:

$$\begin{pmatrix} 1 & 4 & -1 & 3 \\ 2 & 7 & -10 & -7 \\ 0 & 8 & 0 & 5 \end{pmatrix} \in \mathbb{R}^{3x4}$$

$$\begin{pmatrix} 1 & -4 \\ -1 & 3 \\ 3 & 6 \end{pmatrix} \in \mathbb{R}^{3x2}$$

Matrices

A square matrix is a matrix with the same number of rows and columns. In other words, $A \in \mathbb{R}^{nxn}$ for some choice of n.

Examples:

$$\begin{pmatrix} 1 & 3 & 3 \\ 0 & -1 & 0 \\ 5 & 2 & 2 \end{pmatrix} \in \mathbb{R}^{3x3}$$

$$\begin{pmatrix} 0 & 1 \\ 5 & -1 \end{pmatrix} \in \mathbb{R}^{2x2}$$

Basic operations on matrices

Addition: Two matrices $A, B \in \mathbb{R}^{mxn}$ can be added (entrywise).

Examples:

$$\begin{pmatrix} 1 & 4 & -1 & 3 \\ 2 & 7 & -10 & -7 \\ 0 & 8 & 0 & 5 \end{pmatrix} + \begin{pmatrix} 0 & 1 & -1 & 3 \\ 9 & 2 & 0 & 0 \\ 3 & 1 & 3 & 3 \end{pmatrix} =$$

$$\begin{pmatrix} 1 & 5 & -2 & 6 \\ 11 & 9 & -10 & -7 \\ 3 & 9 & 3 & 8 \end{pmatrix}$$

Product of matrices

Multiplication: For two matrices A, B, the product $A \cdot B$ is defined only when: # of columns in A = # of rows in B.

If $A \in \mathbb{R}^{m x \ell}$ and $B \in \mathbb{R}^{\ell x n}$, then $A \cdot B \in \mathbb{R}^{m x n}$.

In particular, $(A \cdot B)_{ij} = \sum_{k=1}^{\ell} A_{ik} \cdot B_{kj}$.

Product of matrices

Example of matrix multiplication.

$$\begin{pmatrix} 3 & 1 & -1 \\ 2 & 4 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 3 \\ 4 & 2 \\ 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \cdot 1 + 1 \cdot 4 + (-1) \cdot 1 & 3 \cdot 3 + 1 \cdot 2 + (-1) \cdot 0 \\ 2 \cdot 1 + 4 \cdot 4 + 0 \cdot 1 & 2 \cdot 3 + 4 \cdot 2 + 0 \cdot 0 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 11 \\ 18 & 14 \end{pmatrix}$$

Non-commutativity of matrix product

Principal source of difficulty in matrix algebra:

Matrix product does not commute.

In other words, for two matrices A, B: $A \cdot B \neq B \cdot A$.

Example: $A = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}, B = \begin{pmatrix} 0 & 6 \\ -1 & -2 \end{pmatrix}$;

$$A \cdot B = \begin{pmatrix} -4 & 4 \\ -2 & 2 \end{pmatrix},\ B \cdot A = \begin{pmatrix} 6 & 12 \\ -4 & -8 \end{pmatrix}.$$

Transpose

For any matrix $A \in \mathbb{R}^{mxn}$, its transpose is $A^t \in \mathbb{R}^{nxm}$ (obtained by switching the rows and columns of A).

Easy facts:

i. $(A^t)^t = A$.

ii. $(A + B)^t = A^t + B^t$.

iii. $(A \cdot B)^t = B^t \cdot A^t$.

Identity matrix

For any $n \geq 1$, the identity matrix in n dimensions $I_n \in \mathbb{R}^{nxn}$ has 1 on the diagonals and 0 on the off-diagonals.

$$I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Determinant of a square matrix

To define $\det(A) \in \mathbb{R}$, we need the notion of minor.

Corresponding to $1 \leq i, j \leq n$, the minor M_{ij} is obtained by deleting the i^{th} row and j^{th} column.

Minors of a square matrix

Example: Suppose $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$.

$$M_{11} = \begin{pmatrix} 5 & 6 \\ 8 & 9 \end{pmatrix}, \qquad M_{23} = \begin{pmatrix} 1 & 2 \\ 7 & 8 \end{pmatrix}, \qquad M_{32} = \begin{pmatrix} 1 & 3 \\ 4 & 6 \end{pmatrix}$$

Determinant of a square matrix

For $A \in \mathbb{R}^{1\text{x}1}$, i.e., it's a single number: $\det(A) = A$.

For $A \in \mathbb{R}^{n\text{x}n}$, $\det(A) = \sum_{j=1}^{n}(-1)^{1+j} \cdot \det(M_{1j})$.

Example: Determinant computation

Consider a simple 2x2 matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$.

$$\det(A) = 1 \cdot 1 \cdot \det(3) + (-1) \cdot 2 \cdot \det(2)$$
$$= 1 \cdot 1 \cdot 3 - 1 \cdot 2 \cdot 2 = -1.$$

More about determinants

Two basic properties of determinants:

1. Multiplicative: $\det(A \cdot B) = \det(A) \cdot \det(B)$.
2. Invariance under transpose: $\det(A^t) = \det(A)$.

More about determinants

Invariance under elementary operations: A crucial property of Determinants is its invariance under *elementary row operations*. If you add a scalar multiple of a row to another, you do not change the determinant.

More about determinants

Invariance under elementary operations

Example: $\begin{pmatrix} 5 & 2 & 1 \\ 3 & -1 & 0 \\ 1 & 7 & 9 \end{pmatrix} \quad \rightarrow_{R_1 \rightarrow R_1+2R_3} \quad \begin{pmatrix} 7 & 16 & 19 \\ 3 & -1 & 0 \\ 1 & 7 & 9 \end{pmatrix}$

determinant = -77

determinant = -77

Computing determinants

- Invariance of determinants under elementary row operations
  $\rightarrow$ transform into a diagonal matrix with the same determinant.

- Easy to compute determinant of diagonal matrices – product of the diagonal entries!

- Efficient algorithm to compute determinants – for $A \in \mathbb{R}^{nxn}$, we can compute determinants in time $O(n^3)$.

Matrix inverse

For a square matrix $A \in \mathbb{R}^{nxn}$, its inverse is the unique matrix B (if one exists), $A \cdot B = B \cdot A = I_n$.

Matrix inverse is not always guaranteed to exist.

The inverse of A (if it exists) is denoted by A^{-1}.

Matrix inverse

Example: $A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$; $A^{-1} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

$$A \cdot A^{-1} = \begin{pmatrix} 1 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix} . \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$A^{-1} \cdot A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Matrix inverse (non-existence)

Not every square matrix $A \in \mathbb{R}^{nxn}$ has an inverse.

Example: $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ is a matrix with no inverse.

Question: When does a matrix have an inverse?

Existence of matrix inverse and determinants

Claim: A matrix $A \in \mathbb{R}^{nxn}$ has an inverse if and only if $\det(A) \neq 0$.

Consistent with previous example: $\det\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 0$ and the inverse of $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ does not exist.

More properties of matrix inverse

Matrix inverses satisfy many useful properties:

I. $(A \cdot B)^{-1} = B^{-1} \cdot A^{-1}$.

II. $(A^t)^{-1} = (A^{-1})^t$.

III. $(A^{-1})^{-1} = A$.

All these are easy proofs. Left as an exercise.

Summary

In summary:

Quick tour of the basics of matrix algebra with various operations like product, inverse, transpose, determinant etc.

Also explored relations between these operations: e.g., matrix A has an inverse if and only if $\det(A) \neq 0$.

Algorithms for Big Data

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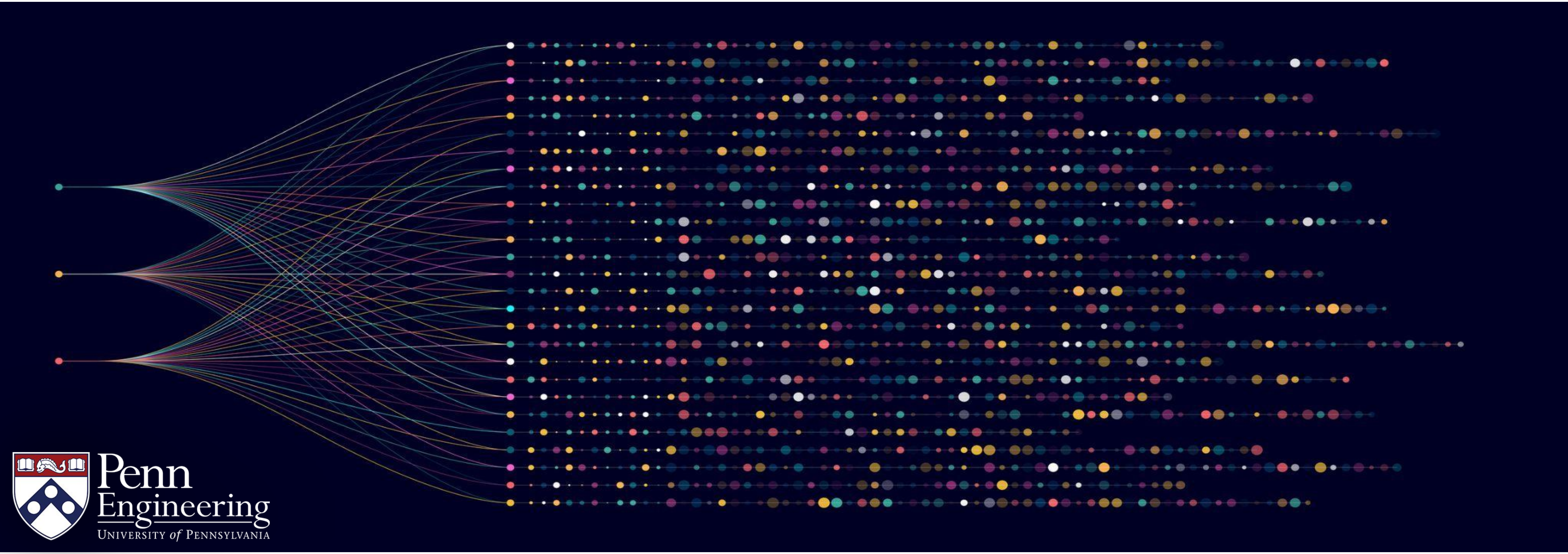
Reasoning in basic linear algebra



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All bases have the same cardinality

Previous video: If B_1 and B_2 are two bases for a subspace V, then B_1 and B_2 have the same number of elements.

We will prove this fact now.

Size of linearly independent sets

Let V be a subspace and $B_1 = \{u^{(1)}, \dots, u^{(m)}\}$ be a basis.

Let $A = \{v^{(1)}, \dots, v^{(\ell)}\}$ with $\ell > m$.

Claim: A is linearly dependent.

Size of linearly independent sets

Let V be a subspace and $B_1 = \{u^{(1)}, \dots, u^{(m)}\}$ be a basis.

Let $A = \{v^{(1)}, \dots, v^{(\ell)}\}$ with $\ell > m$.

Claim: A is linearly dependent.

Suppose B_1 and B_2 are two bases. As B_2 is linearly independent, $|B_2| \leq |B_1|$. Likewise, $|B_1| \leq |B_2|$. Thus, $|B_2| = |B_1|$.

Size of linearly independent sets

Let V be a subspace and $B_1 = \{u^{(1)}, \dots, u^{(m)}\}$ be a basis.

Let $A = \{v^{(1)}, \dots, v^{(\ell)}\}$ with $\ell > m$.

Claim: A is linearly dependent.

Representing each element in A in terms of B_1.

Since B_1 is a basis, for each element $v^{(j)} \in A$, we can write:

$v^{(j)} = \sum_{i=1}^{m} \beta_{ij} u^{(i)}$ for $\beta_{ij} \in \mathbb{R} - (1)$.

We will now show that $v^{(1)}, \ldots, v^{(\ell)}$ are linearly dependent.

Key idea

Use: rank nullity theorem

Consider a system of m equations in ℓ variables – call the variables $z_1, \ldots, z_\ell$ and for each $1 \leq i \leq m$, we have the equation

$$\sum_{j=1}^{\ell} \beta_{ij} \; z_j = 0.$$

Key idea

Use: rank nullity theorem

Consider a system of m equations in ℓ variables – call the variables $z_1, \ldots, z_\ell$ and for each $1 \leq i \leq m$, we have the equation

$$\sum_{j=1}^{\ell} \beta_{ij} \; z_j = 0.$$

Homogenous system with m equations and ℓ variables and $m < \ell$.

Implication of the rank nullity theorem

Rank nullity theorem implies a non zero solution.

There exists $z_1^*, \dots, z_\ell^*$ (not all zero) such that for all $1 \le i \le m$,

$$\sum_{j=1}^{\ell} \beta_{ij}\ z_j^* = 0\ -(2).$$

We will now use $z_1^*, \dots, z_\ell^*$ to show that the vectors in A are linearly dependent.

Combining our observations

Recall (1) –for every $v^{(j)} \in A$, $v^{(j)} = \sum_{i=1}^{m} \beta_{ij} u^{(i)}$.

(2) –for every $1 \leq i \leq m$, $\sum_{j=1}^{\ell} \beta_{ij}\ z_j^* = 0$.

Thus, $\sum_{j=1}^{\ell} z_j^* v^{(j)} = \sum_{j=1}^{\ell} z_j^* \sum_{i=1}^{m} \beta_{ij} u^{(i)}$ (by (1))

$= \sum_{i=1}^{m} u^{(i)} \left(\sum_{j=1}^{\ell} \beta_{ij} z_j^* \right)$ (rearranging i, j)

$= \mathbf{0}$ (using (2)).

Implication: linear dependence of A.

$$\begin{aligned}
\sum_{j=1}^{\ell} z_j^* v^{(j)} &= \sum_{j=1}^{\ell} z_j^* \sum_{i=1}^{m} \beta_{ij} u^{(i)} \quad \text{(by (1))} \\
&= \sum_{i=1}^{m} u^{(i)} \left(\sum_{j=1}^{\ell} \beta_{ij} z_j^* \right) \quad \text{(rearranging } i, j\text{)} \\
&= \mathbf{0} \quad \text{(using (2)).}
\end{aligned}$$

As $\{z_j^*\}$ are not all zero, it follows that elements of A are linearly dependent. Thus, A cannot be a basis.

Recap of the what we proved

Thus, we have established that if we have a basis B_1 with $|B_1| = m$, any set A such that $|A| = \ell > m$ must be linearly dependent.

This also proves that two bases B_1 and B_2 have the same size.

Summary

Reasoning about the solutions of a linear system and rank-nullity theorem in particular – elementary but powerful tool in linear algebra.

Applications in recitation and assignments.